Introduction to/ Overview of Metabolism

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Objectives:

At the end of this lecture, it is expected to:

- Describe the concepts of metabolism and the types of metabolic pathways– Anabolic, Catabolic and amphibolic pathways
- Understand the differences between Anabolism and Catabolism
- State the major functions of metabolism
- Describe the general principles common to metabolic pathways
- Describe the stages of Metabolism
- Explain the Regulation of Metabolisms
- Explain the significance of metabolism in medicine and allied medical sciences
- Describe the mechanism of energy Coupling
- Explain the role of ATP in chemical reactions of our body

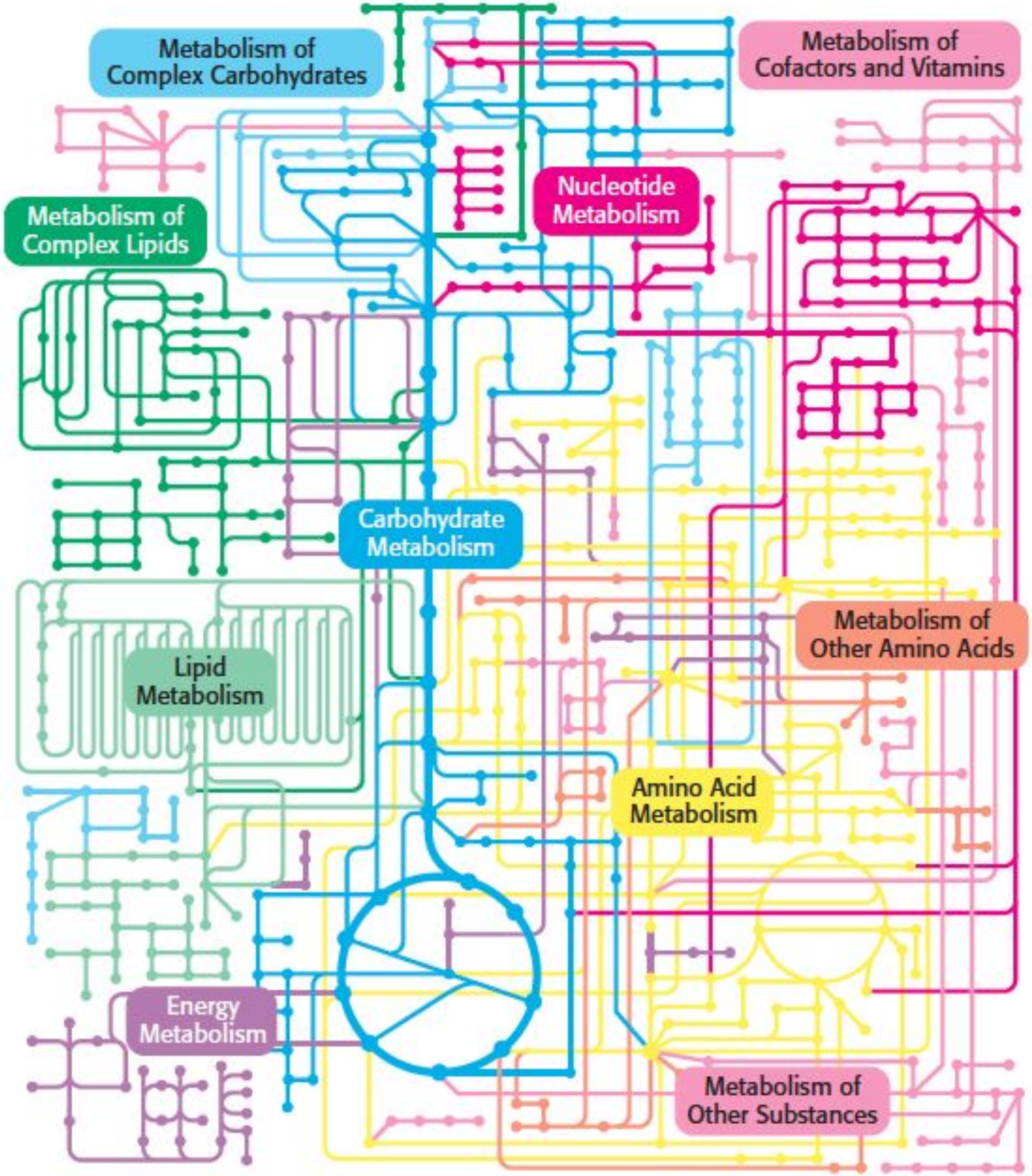
Metabolism

- Total sum of the chemical reaction happening in a living organism
- Highly coordinated and purposeful activity
- Includes:
  - Anabolism:
    - Endergonic; Energy-requiring process where small molecules joined to form larger molecules
    - E.g. Glycogenesis
  - Catabolism:
    - Exergonic; Energy-releasing process where large molecules broken down to smaller
    - E.g. Glycogenolysis

Metabolic Pathways

- A particular set of reactions that carries out a certain function or functions.
- Occurs in small discrete steps, each of which is catalyzed by an enzyme
- Types:
  - Anabolic Pathways
  - Catabolic Pathways
  - Amphibolic Pathways

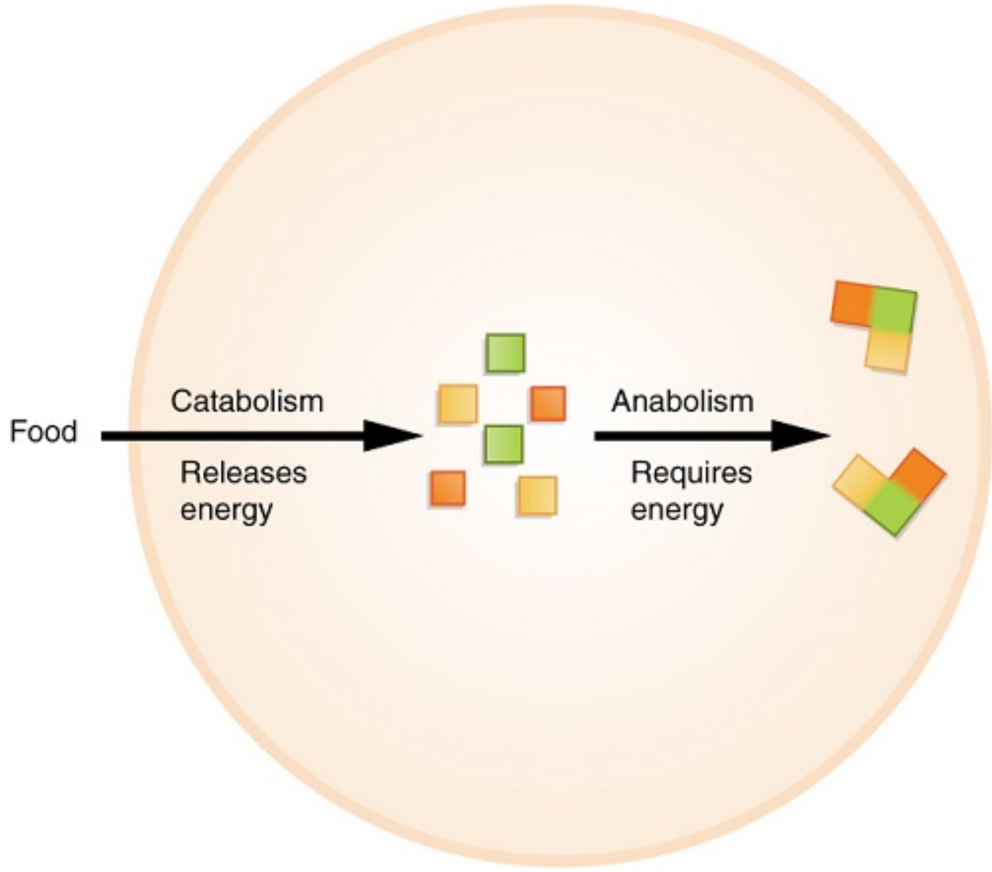
Metabolic pathways



Amphibolic Pathways

- Amphi = Dual, amphibolic: dual pathway (both catabolic and anabolic)
- For example: Krebs cycle
  - mainly a catabolic cycle, but with some anabolic features,
  - e.g., part of Krebs cycle is used for the synthesis of glucose from amino acids

Anabolism Vs Catabolism



Overview of Metabolism

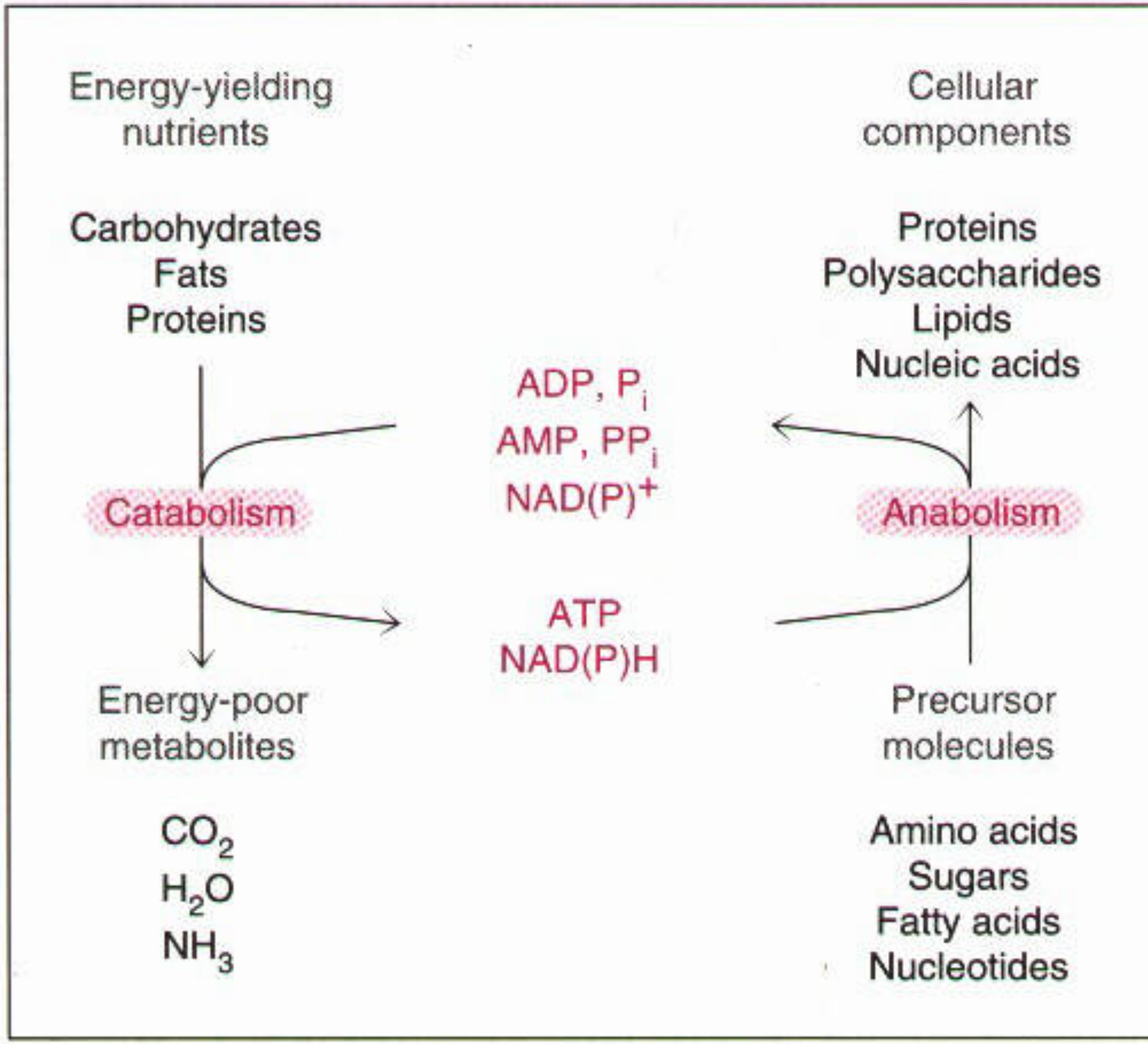


FIGURE 7–1. Overview of metabolism. *The exergonic reactions of catabolism sustain ATP, NADH, and NADPH formation. ATP and NADPH provide energy and reducing equivalents for the endergonic reactions of anabolism.*

Differences b/w Anabolism and Catabolism

	CATABOLISM	ANABOLISM
Energy	– Releases ATP energy – Potential energy converted into kinetic energy	– Requires ATP energy – Kinetic energy converted into potential energy
Reaction type	Exergonic	Endergonic
Hormones	Adrenaline, Glucagon, Cytokines, Cortisol	Estrogen, Testosterone, Growth hormone, Insulin
Importance	Provides energy for anabolism	– Heats the body – Enables muscle contraction – Supports new cell growth – Supports storage of energy – Body tissue maintenance
Oxygen	Utilizes oxygen	Does not utilize oxygen
Effects on exercise	Catabolic exercises are usually aerobic and are good at burning calories and fat	Anabolic exercises, often anaerobic in nature and are generally muscle mass building
Examples	• Cell respiration • Digestion • Excretion	• Assimilation in animals • Photosynthesis in plants

Intermediary Metabolism

- Biochemical reactions that degrade, synthesize, or interconvert small molecules inside living cells.
- Connect with metabolism of most biomolecules
  - e.g. TCA cycle

Metabolic Intermediates

- Organic compounds that comprising the biochemical pathways
- Contain carbon, hydrogen, and oxygen.
- Some also contain nitrogen or sulfur.
- In most instances, have no function
  - but in some cases act as key regulators of other pathways (e.g. citric acid)

Homeostasis

- An organism's tendency or drive to maintain the normalcy of its internal environment,
  - including maintaining the concentration of nutrients and metabolites within relatively strict limits.
  - e.g.: Glucose homeostasis

WHAT DO METABOLIC PATHWAYS ACCOMPLISH?

- The primary functions of metabolism are:

  a. **Generation & Utilization of energy**

     - Energy in carbohydrates, lipids, proteins is used to produce ATP through oxidation-reduction reactions

  b. **Storage of Energy**

  c. **Synthesis of molecules**

     - for cell structure and functioning (i.e. proteins, nucleic acids, lipids, & CHO

  d. **Degradation or Catabolism of Organic Molecules**

  e. **Removal of waste** products

GENERAL PRINCIPLES COMMON TO METABOLIC PATHWAYS

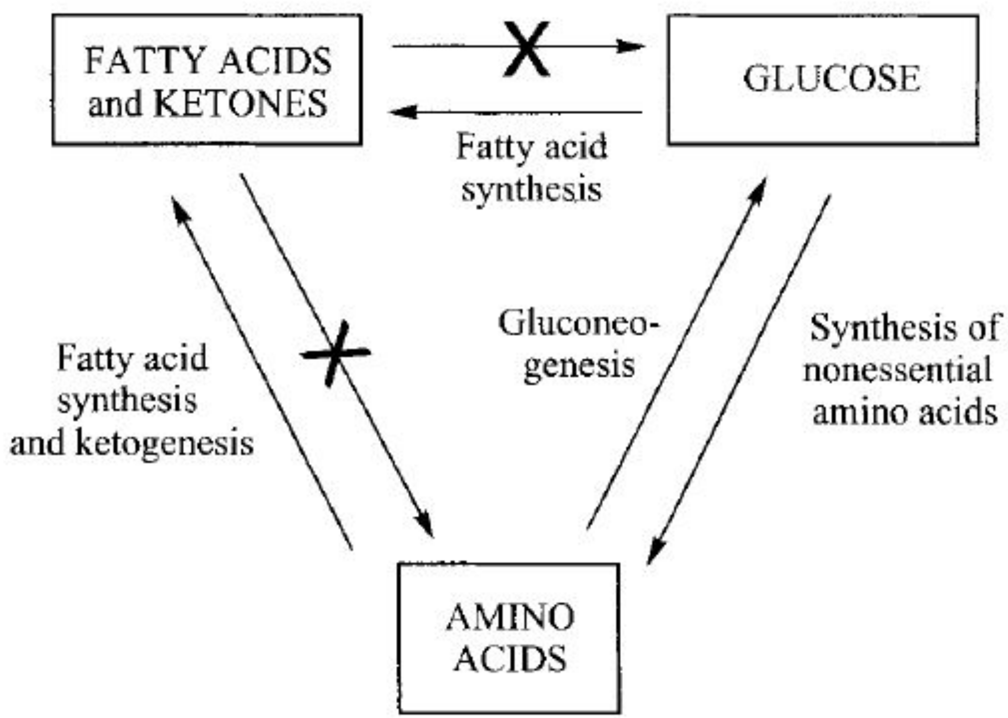
- ATP Provides Energy for Synthesis
- Many Metabolic Reactions Involve Oxidation or Reduction

Types of chemical reactions in metabolism

Type of reaction	Description
Oxidation–reduction	Electron transfer
Ligation requiring ATP cleavage	Formation of covalent bonds (i.e., carbon–carbon bonds)
Isomerization	Rearrangement of atoms to form isomers
Group transfer	Transfer of a functional group from one molecule to another
Hydrolytic	Cleavage of bonds by the addition of water
Addition or removal of functional groups	Addition of functional groups to double bonds or their removal to form double bonds

GENERAL PRINCIPLES COMMON TO METABOLIC PATHWAYS (Cont'd)

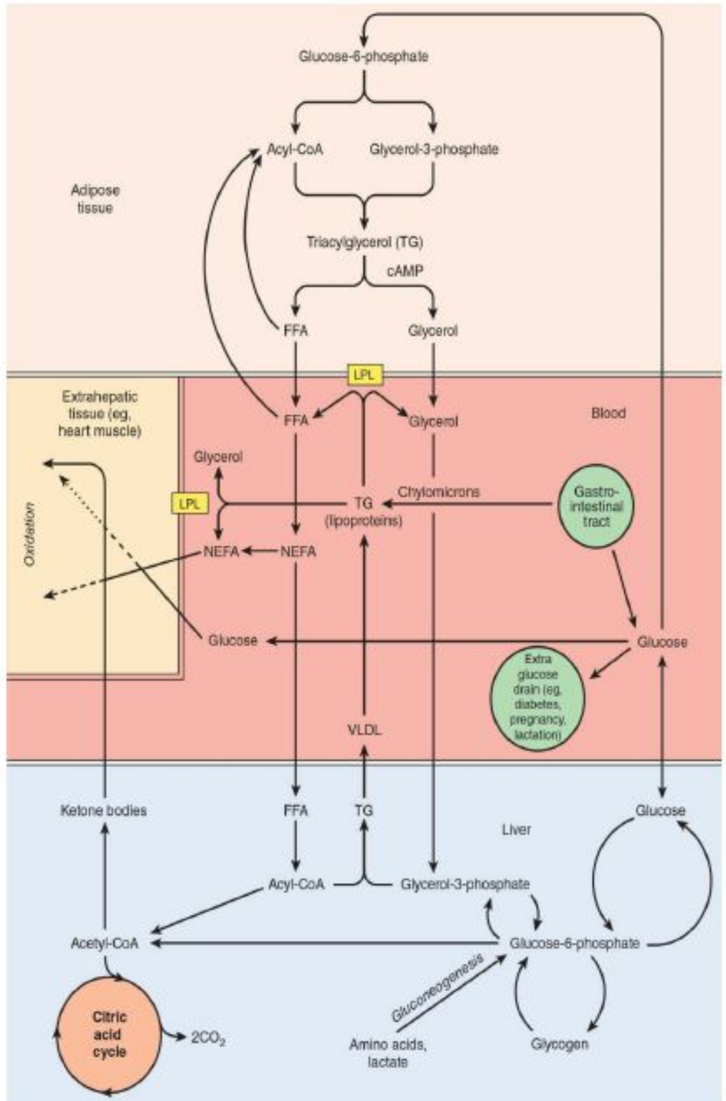
- Only Certain Metabolic Reactions Occur in Human Metabolism



GENERAL PRINCIPLES COMMON TO METABOLIC PATHWAYS (Cont'd)

- Some Organic Molecules are Nutritionally Essential to Human Health
- Some are Irreversible or Contain Irreversible Steps
- Not Necessarily Linear
- All are Regulated
- All are Interconnected (see next slide)

Metabolic interrelationships among adipose tissue, the liver, and extrahepatic tissues.



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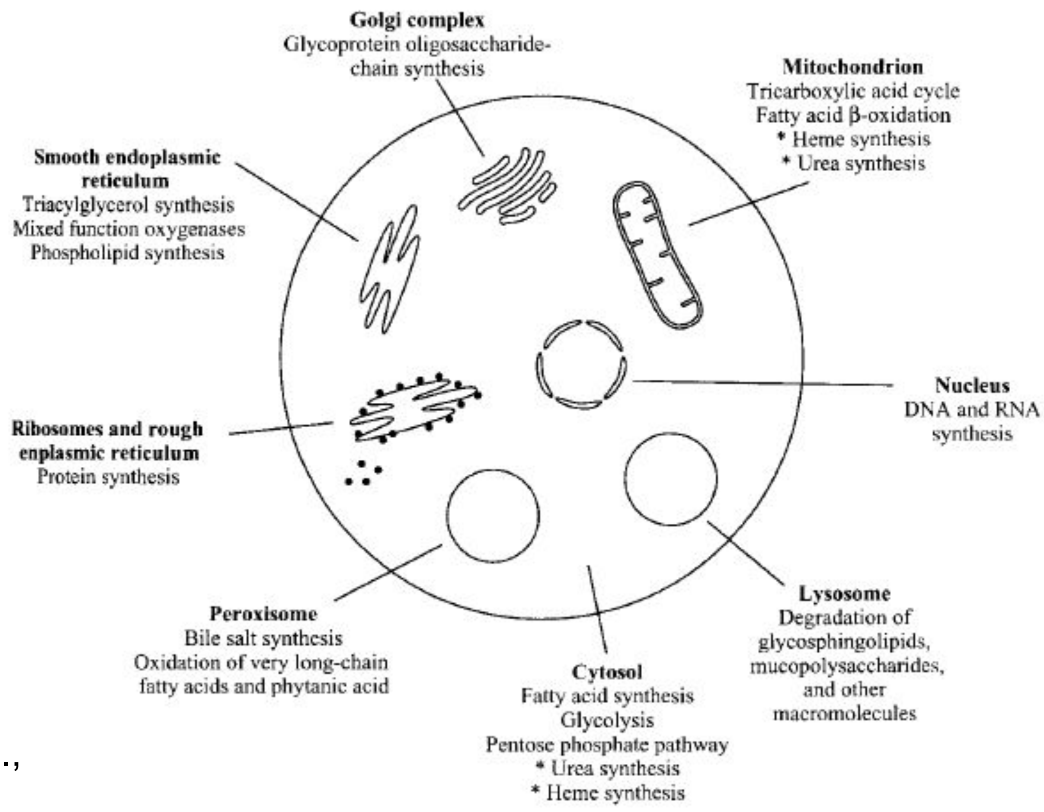
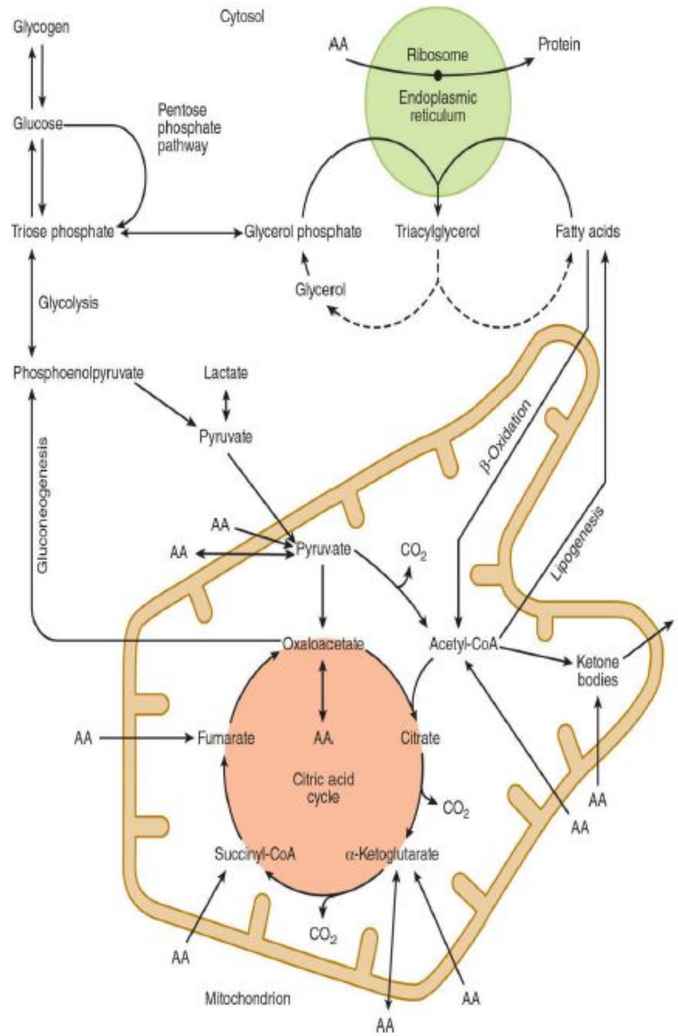
GENERAL PRINCIPLES COMMON TO METABOLIC PATHWAYS (Cont'd)

- A Different Repertoire of Pathways Occurs in Different Organs, e.g.: Urea cycle in liver.
- Localized to Specific Compartments Within the Cell



Intracellular location and overview of major metabolic pathways in a liver parenchymal cell.



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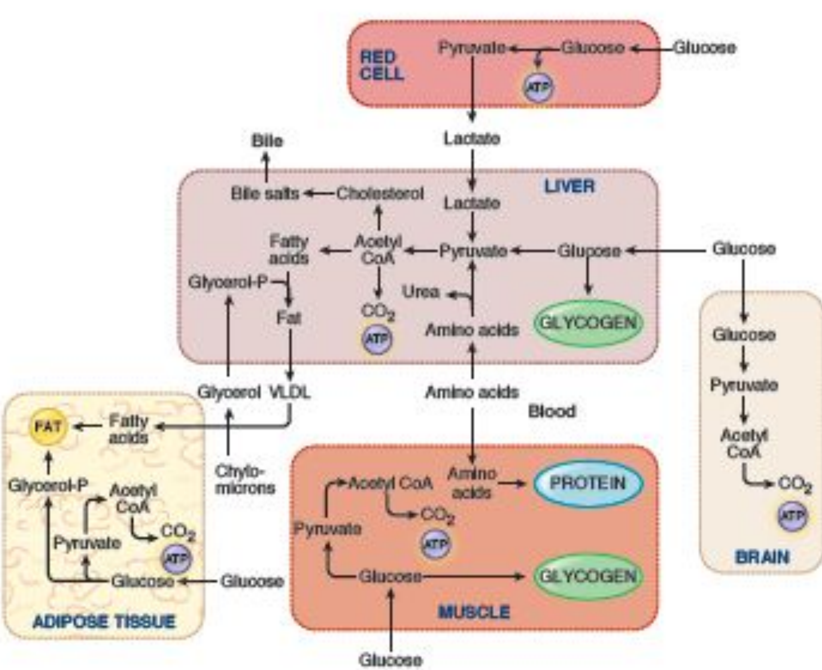
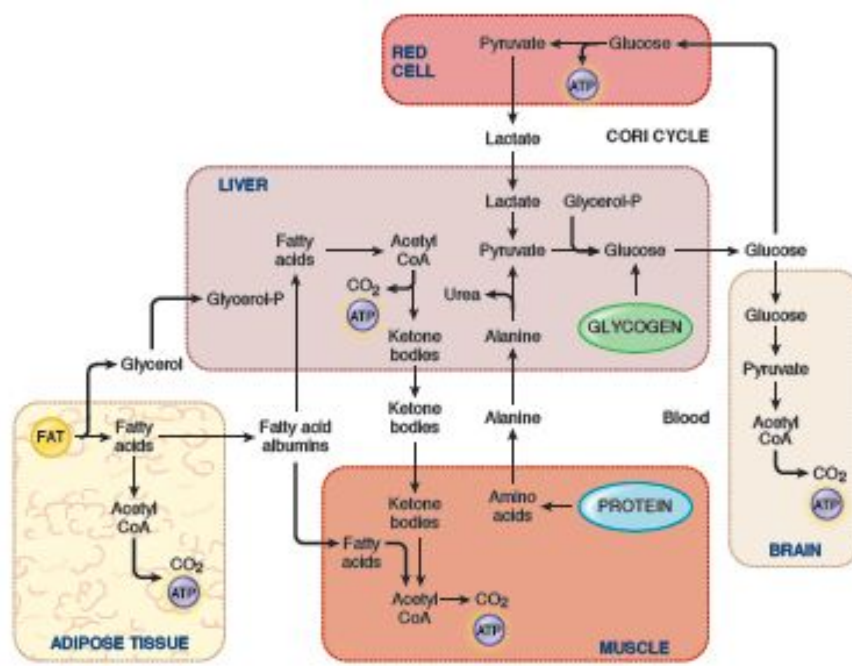
GENERAL PRINCIPLES COMMON TO METABOLIC PATHWAYS (Cont'd)

- Metabolic Processes Occur in the Fed State and Fasting are different.



(Fed State)



(Fasting State)

Some activated carriers in metabolism

Carrier molecule in activated form	Group carried	Vitamin precursor
ATP	Phosphoryl	
NADH and NADPH	Electrons	Nicotinate (niacin)
$FADH_2$	Electrons	Riboflavin (vitamin B_2)
$FMNH_2$	Electrons	Riboflavin (vitamin B_2)
Coenzyme A	Acyl	Pantothenate
Lipoamide	Acyl	
Thiamine pyrophosphate	Aldehyde	Thiamine (vitamin B_1)
Biotin	CO_2	Biotin
Tetrahydrofolate	One-carbon units	Folate
S-Adenosylmethionine	Methyl	
Uridine diphosphate glucose	Glucose	
Cytidine diphosphate diacylglycerol	Phosphatidate	
Nucleoside triphosphates	Nucleotides	

NOTE: Many of the activated carriers are coenzymes that are derived from water-soluble vitamins.

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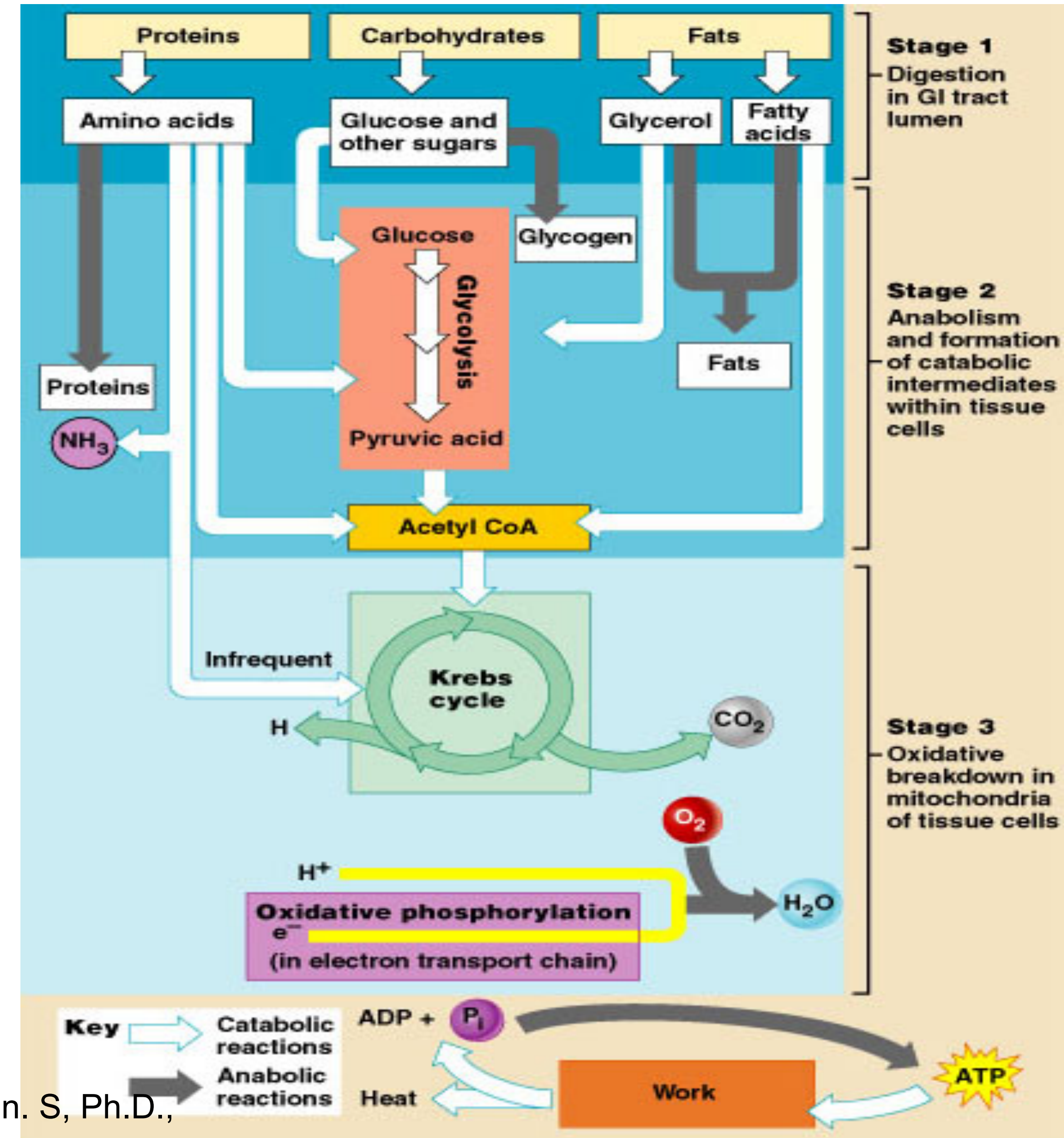
Vitamin	Coenzyme	Typical reaction type	Consequences of deficiency
Thiamine (B_1)	Thiamine pyrophosphate	Aldehyde transfer	Beriberi (weight loss, heart problems, neurological dysfunction)
Riboflavin (B_2)	Flavin adenine dinucleotide (FAD)	Oxidation–reduction	Cheliosis and angular stomatitis (lesions of the mouth), dermatitis
Pyridoxine (B_6)	Pyridoxal phosphate	Group transfer to or from amino acids	Depression, confusion, convulsions
Nicotinic acid (niacin)	Nicotinamide adenine dinucleotide (NAD^+)	Oxidation–reduction	Pellagra (dermatitis, depression, diarrhea)
Pantothenic acid	Coenzyme A	Acyl-group transfer	Hypertension
Biotin	Biotin–lysine adducts (biocytin)	ATP-dependent carboxylation and carboxyl-group transfer	Rash about the eyebrows, muscle pain, fatigue (rare)
Folic acid	Tetrahydrofolate	Transfer of one-carbon components; thymine synthesis	Anemia, neural-tube defects in development
B_{12}	5′-Deoxyadenosyl cobalamin	Transfer of methyl groups; intramolecular rearrangements	Anemia, pernicious anemia, methylmalonic acidosis

Stages of Metabolism

Energy-containing nutrients are processed in three major stages:

1. **Digestion**
2. **Anabolism and formation of catabolic intermediates** (Catabolism)
3. **Oxidative breakdown**



Regulation of Metabolism

The different mechanism of regulation are:

(1) Substrate availability through control of transport across membranes;

(2) Allosteric regulation:

- Enzyme activation by upstream metabolites in the metabolic pathway and/or enzyme inhibition by the product of downstream metabolites in the metabolic pathway

(3) Covalent Modification:

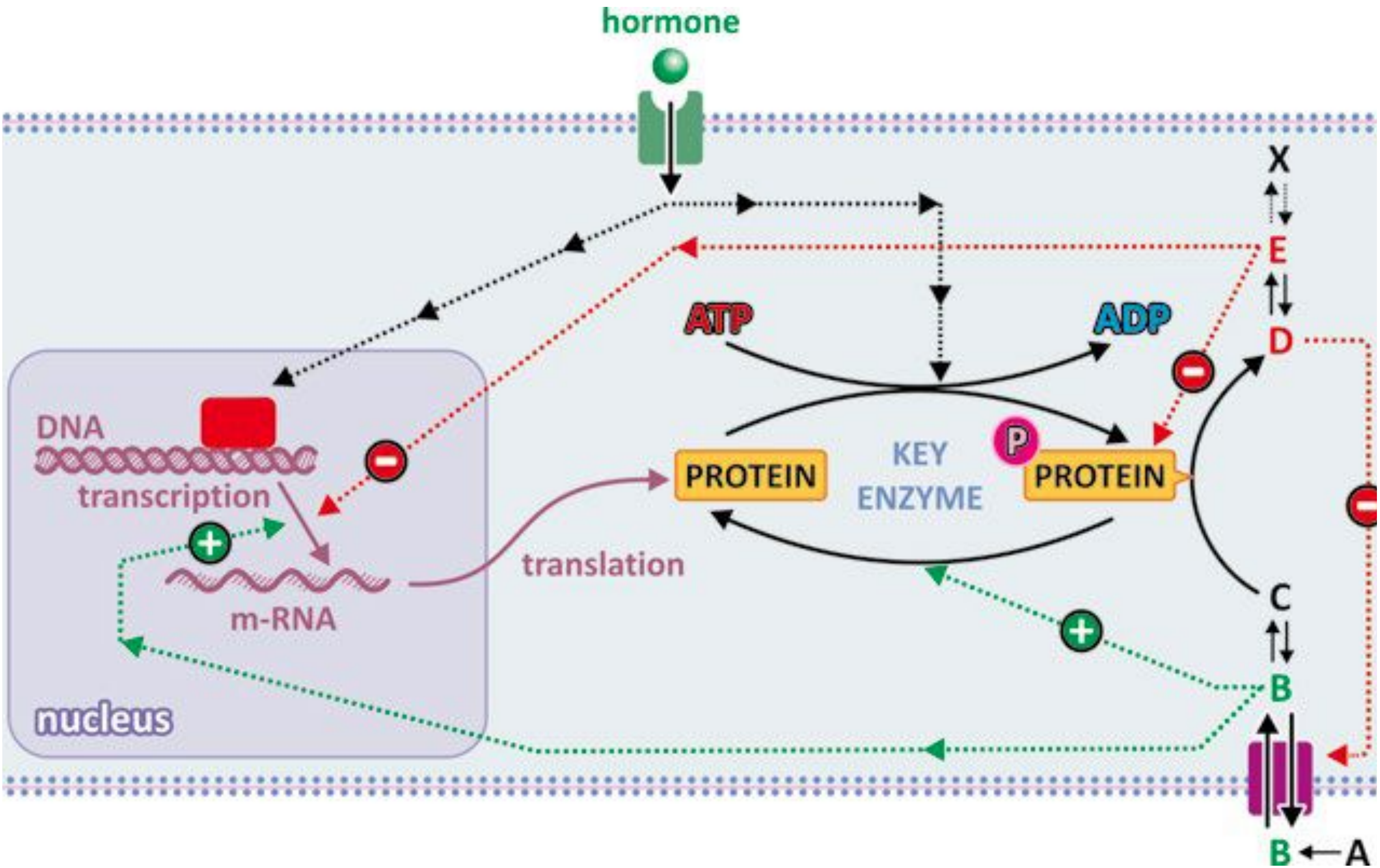
- Activation or inactivation of enzymes by covalent attachment of a phosphate group (P), caused by the binding of a hormone to its receptor (signal transduction);

(4) Regulation of enzymes' gene expression

- Through signal transduction inside cells controlled by hormones.

N.B.: Hormone-dependent mechanisms are slower; the direct effect of metabolites on enzymes (activation or inhibition) is faster because enzymes coexist with regulatory metabolites (named effectors) in the same cell compartment

Regulation of Metabolism



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Significance of Metabolism in Medicine

- To keeps cells and organisms alive
- To screen the diseases
  - Genetic defects of metabolic enzymes - warrant the routine screening of newborns
- To understand the disorders of metabolic - genetic enzyme defect:
- To understand the clinical manifestations of disease
- To identify the strategies for proper diagnosis and treatment
- To understand the metabolic diseases
- To know the targets for drug therapy.
- Antimicrobial drugs often target enzymes in vital metabolic pathways of bacteria and parasites.
- From all this, it should be quite clear that metabolism matters to all those who pursue a career in medicine or a related field.

BIOENERGETICS

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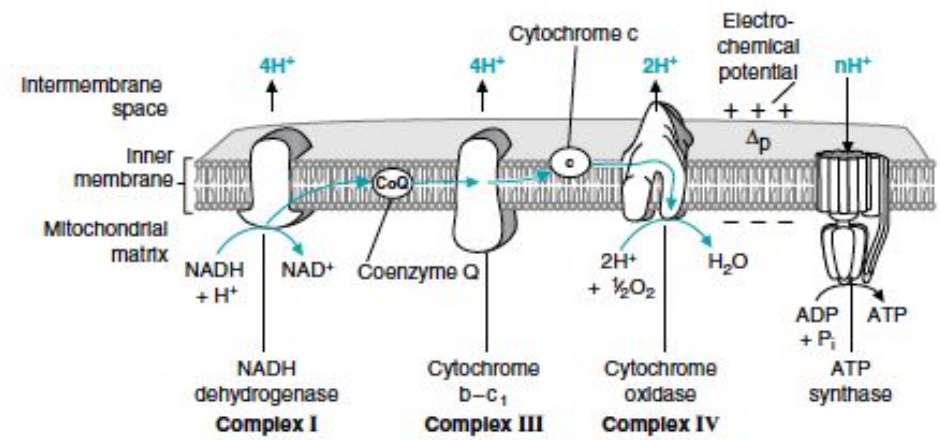
What is Bioenergetics?

- Study of energy transformations in living systems and the organisms that utilize them
- The quantitative study of;
  - the energy transductions that occur in living cells, and
  - the nature and function of the chemical processes underlying these transductions.
- Energy = ability to do work
- Energy is required by all organisms
- Energy may be Kinetic or Potential energy
- Governed by Laws of thermodynamics

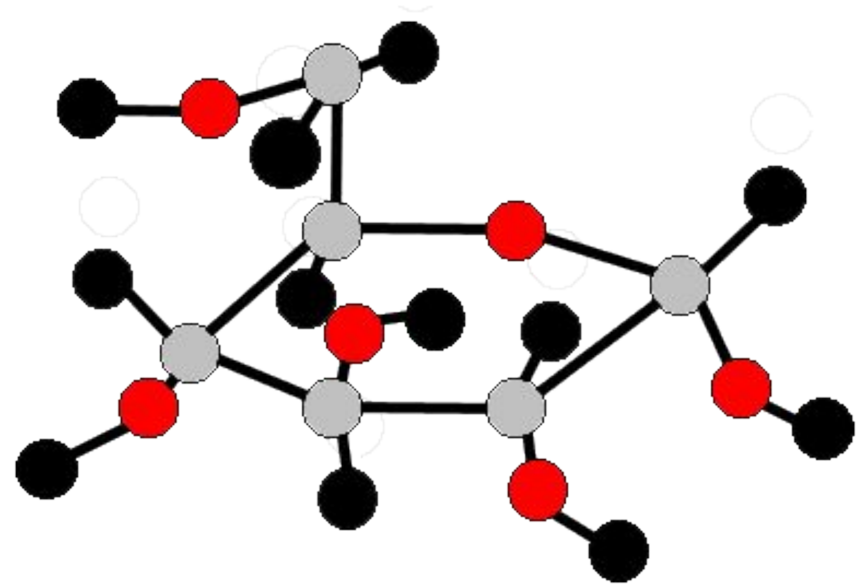
Kinetic Energy

- Energy of Motion



Potential Energy

- Energy of position
- Includes energy stored in chemical bonds



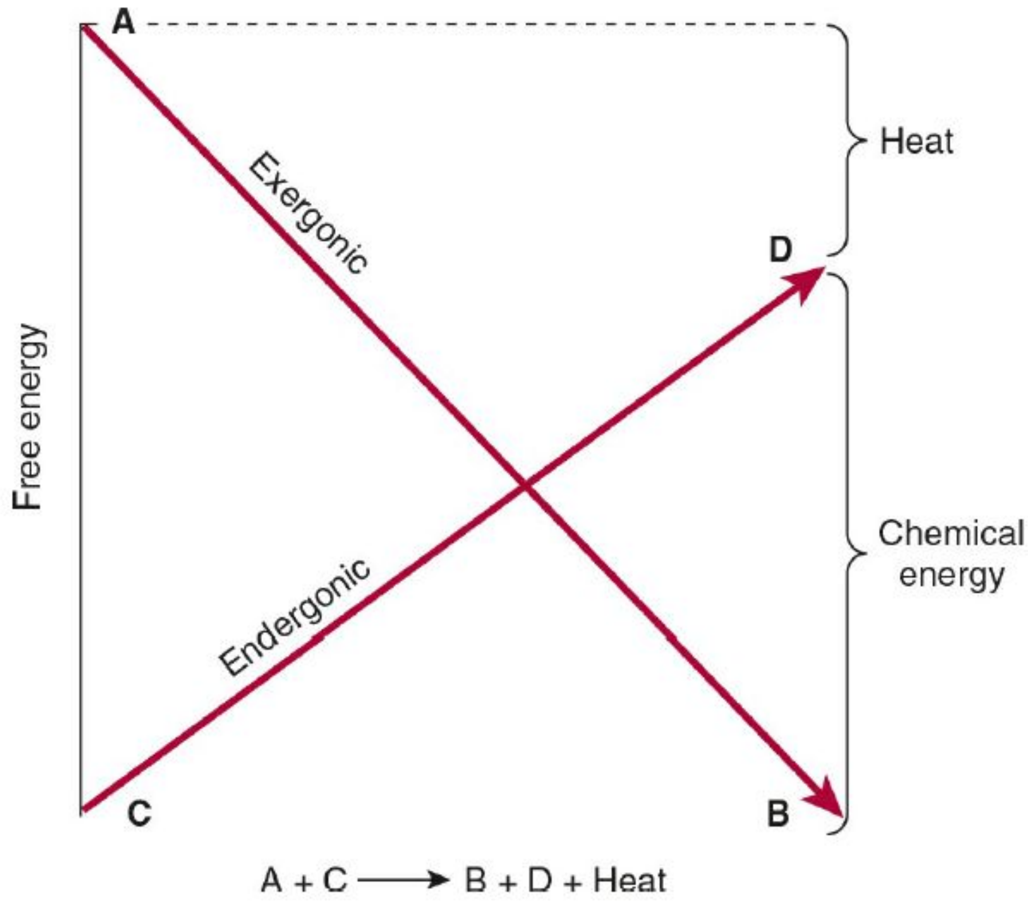
Cellular work

- 3 main kinds:
  - Mechanical - muscle contractions
  - Transport - pumping across membranes
  - Chemical - making polymers
- Require energy
  - All cellular work is powered by ATP

Cell Energy

- Couples an exergonic process to drive an endergonic one.
- ATP is used to couple the reactions together.

Coupling of an exergonic to an endergonic reaction



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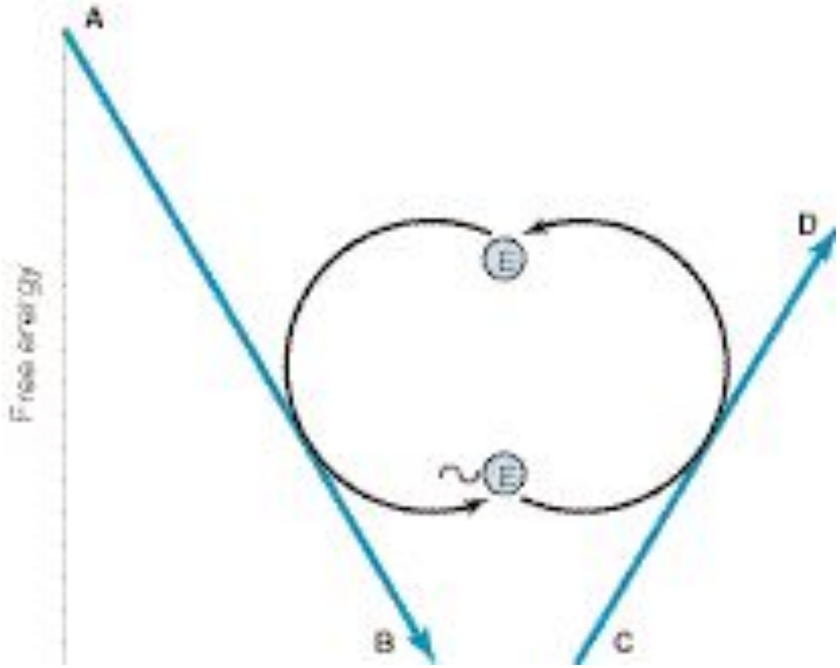
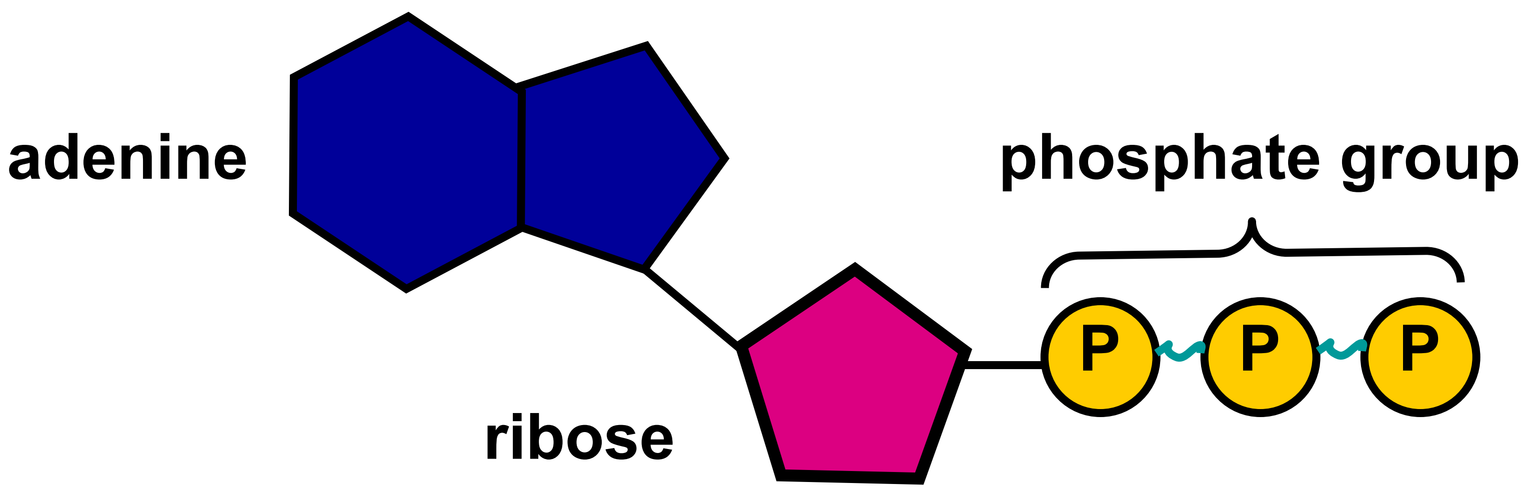
Coupling of an exergonic to an endergonic reaction



- Transfer of free energy from an exergonic to an endergonic reaction via a high-energy intermediate compound (~E), mostly ATP.

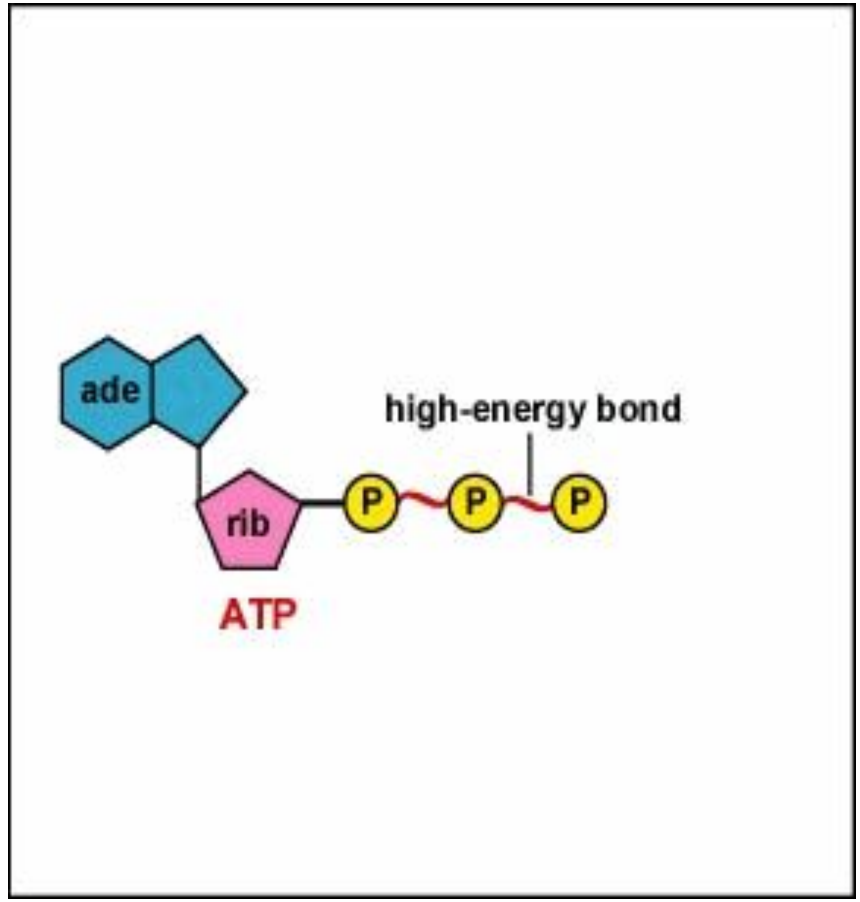
ATP

- Components:
  1. adenine □ nitrogenous base
  2. ribose □ five carbon sugar
  3. phosphate group □ chain of 3

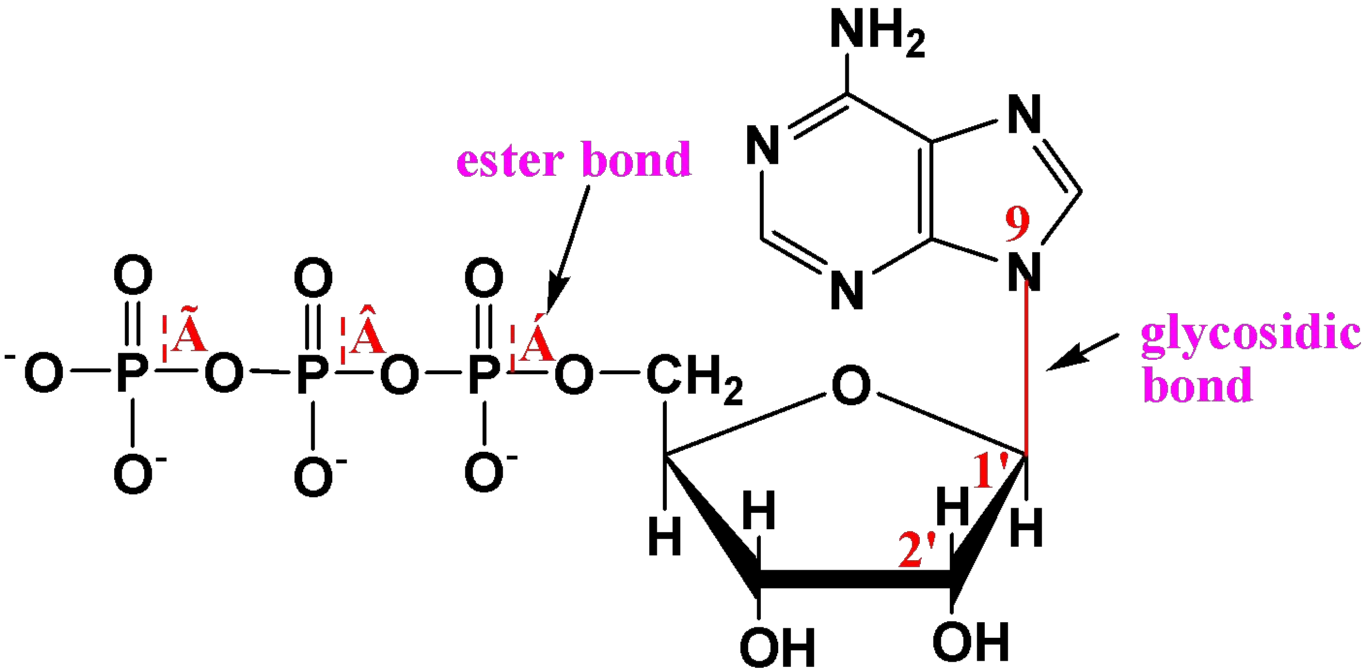


Adenosine Triphosphate

- Three phosphate groups-(two with high energy bonds
- Last phosphate group (PO_4) contains the MOST energy

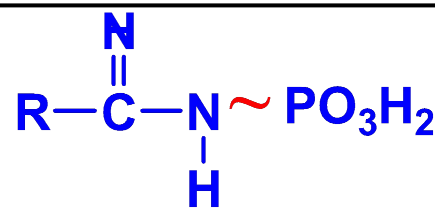
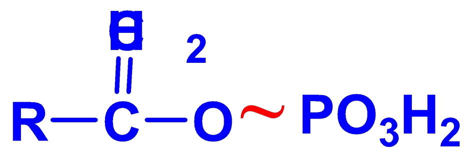
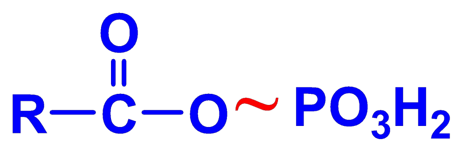
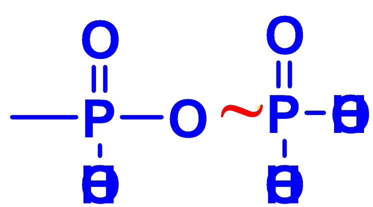
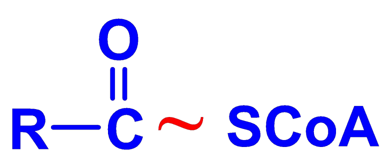


Adenosine triphosphate (ATP)



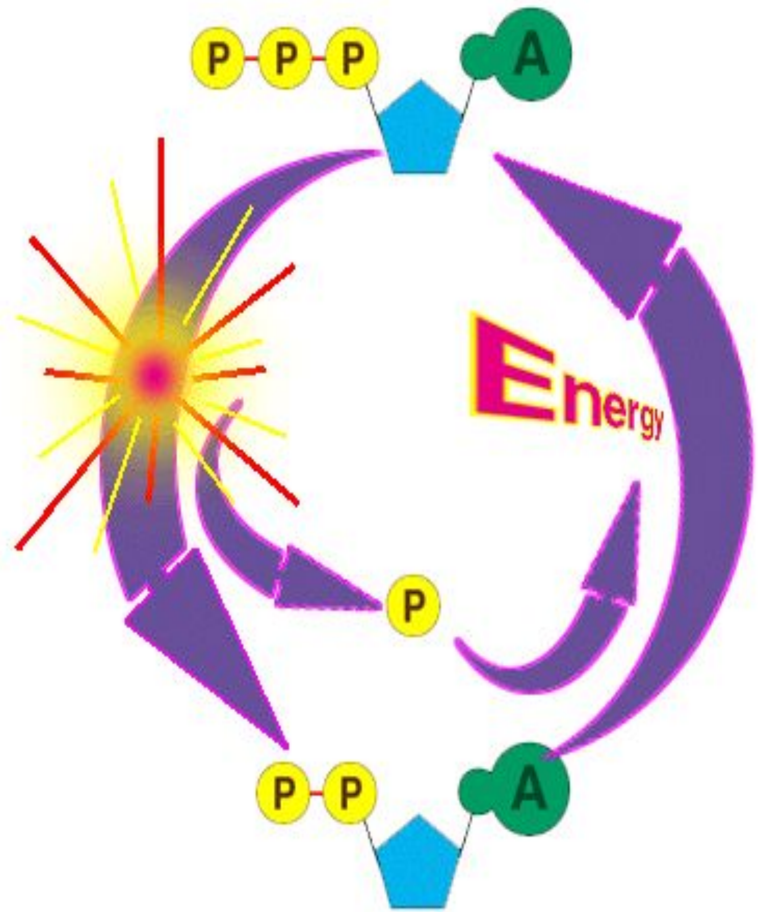
- α bond △ $G^{0\prime}$ = －14.3KJ
- β bond △ $G^{0\prime}$ = －32.2KJ
- γ bond △ $G^{0\prime}$ = －30.5KJ
- >30KJ （ 7kcal ）-high energy bonds

Examples of high-energy compounds

Type of bond	General formula	Example	$\Delta G^{0'}$ kJ/mol
Phospho-guanidines		Creatine phosphate	43.9
Enol phosphates		PEP	61.9
Phosphoric-carboxylic		1,3-bisphospho-glycerate	41.8
Phosphoric acid anhydrides		ATP, GTP, UTP, CTP	30 .5
Thiol esters		Acetyl CoA, Acyl CoA	31.4

Breaking the Bonds of ATP

- Process is called phosphorylation
- Occurs continually in cells
- Enzyme ATP-ase can weaken & break last PO_4 bond releasing energy & free PO_4



Breaking the Bonds of ATP

Potentially, 2 ~P bonds can be cleaved, as 2 phosphates are released by hydrolysis from ATP.

AMP~P~P □ AMP~P + P_i

(ATP □ ADP + P_i)

AMP~P~P □ AMP + P~P

(ATP □ AMP + PP_i)

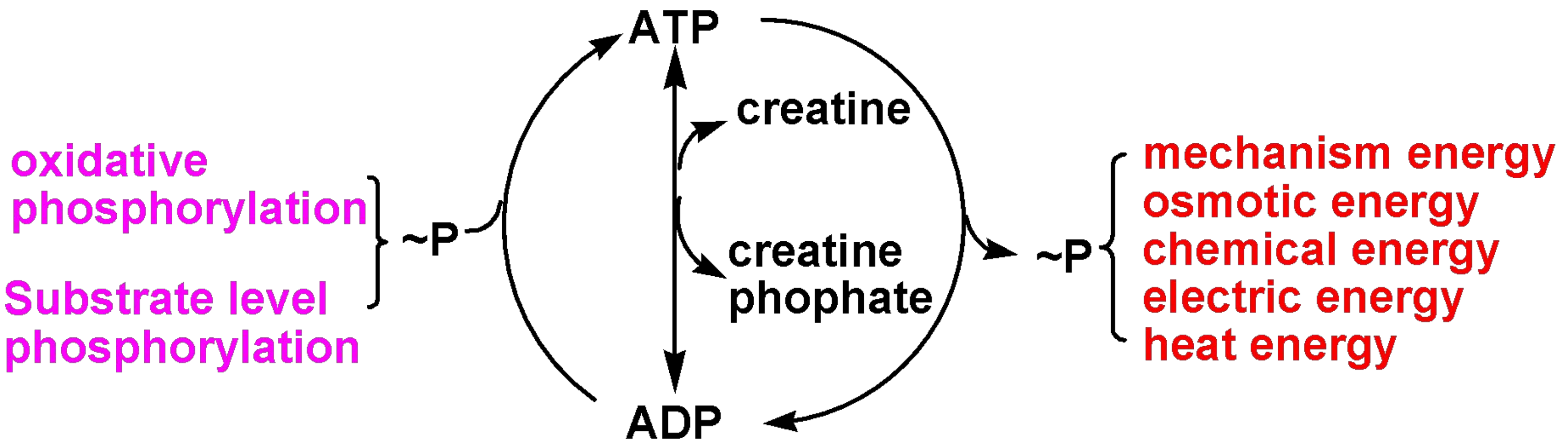
P~P □ 2 P_i

(PP_i □ 2P_i)

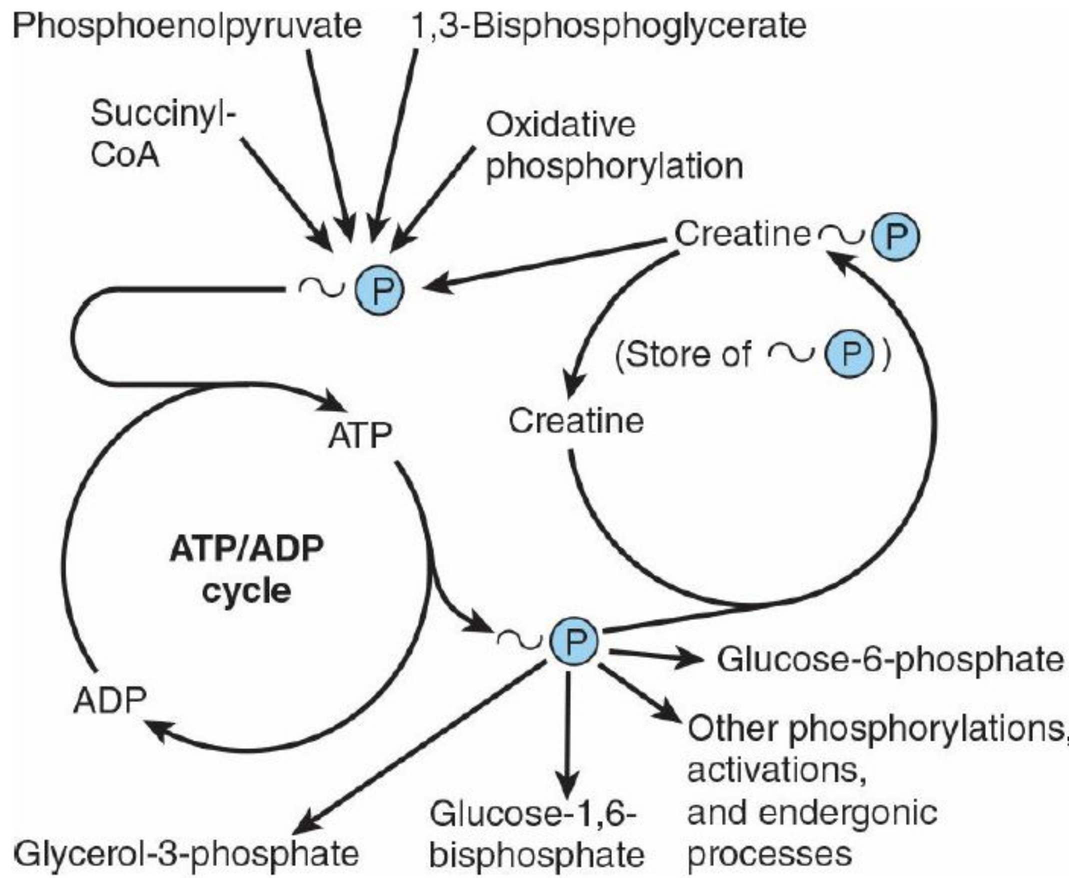
How does ATP work ?

- Organisms use enzymes to break down energy-rich glucose to release its potential energy
- This energy is trapped and stored in the form of adenosine triphosphate(ATP)

Production and application of ATP

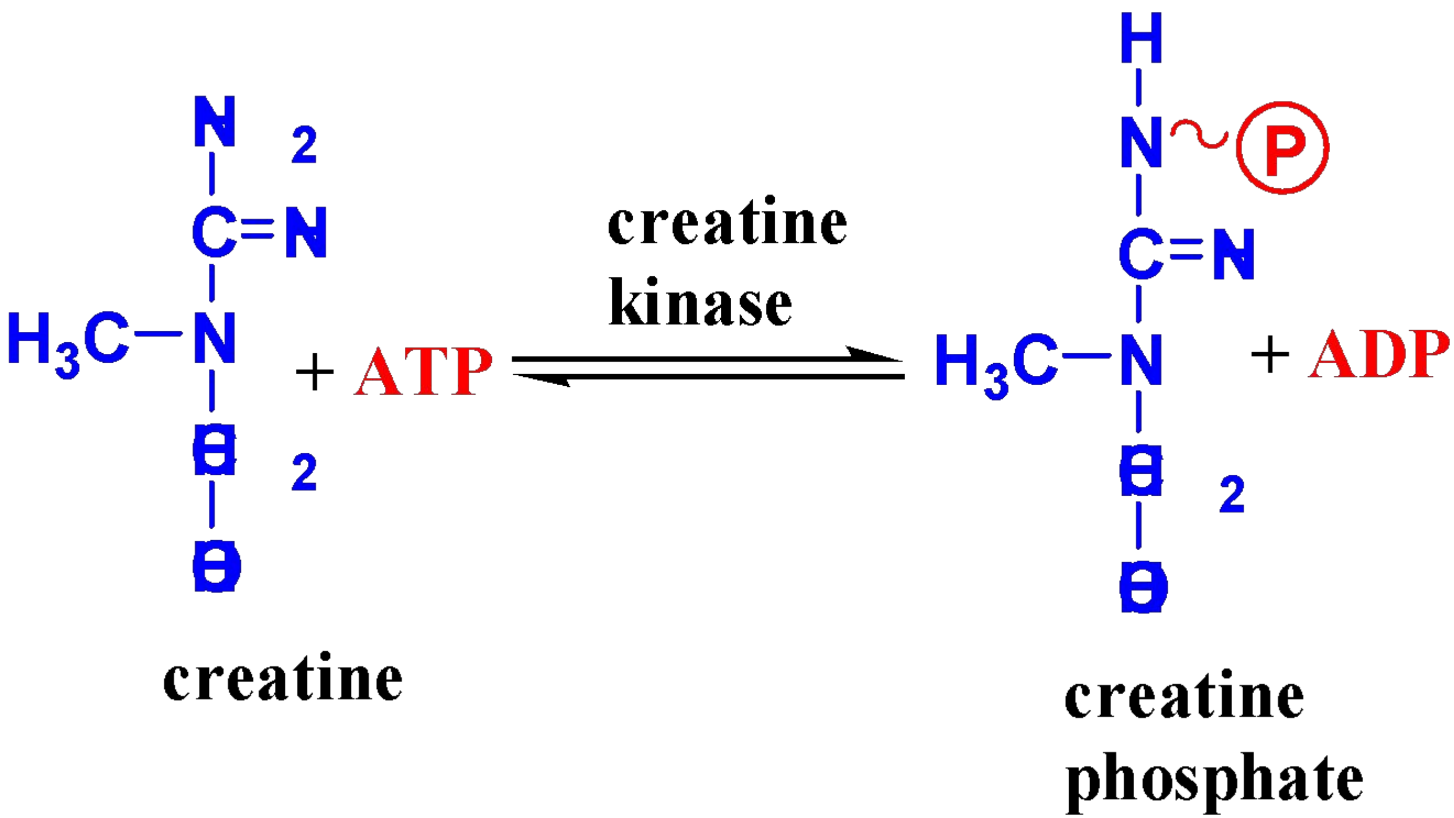


Role of ATP/ADP cycle in transfer of high-energy phosphate

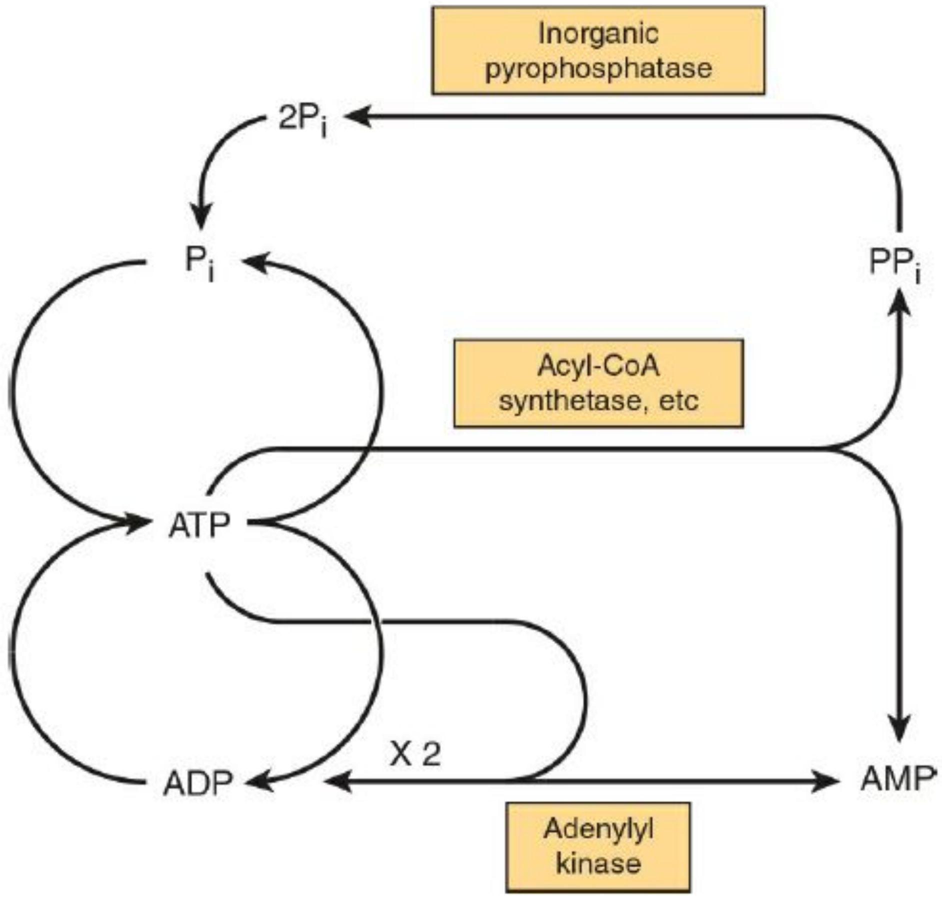


Storage of high-energy phosphate

Phosphocreatine is used in nerve and muscle for storage of ~P bonds.



Phosphate cycles and interchange of adenine nucleotides



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Interchange of adenine nucleotides

- Adenylate Kinase:

  $2\ ADP \leftrightarrow ATP + AMP$

- Nucleoside Diphosphate Kinase catalyzes reversible reactions such as:

  $ATP + GDP \leftrightarrow ADP + GTP$,

  $ATP + UDP \leftrightarrow ADP + UTP$, etc.

How Much ATP Do Cells Use?

- Each cell will generate and consume approximately 10,000,000 molecules of ATP per second

ATP in Cells

- A cell's ATP content is recycled every minute.
- Humans use close to their body weight in ATP daily.
- No ATP production equals quick death.

Read more…

THANK YOU…